

Interim Progress Report

EECS 4312 Project: Requirement Verification for Smart Health & Wellness Center Automation

Date: July 18, 2025

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1. Work Completed So Far

Task 1: Requirement Extraction Using NLP Tools

- **Kanwarjot:**
 - Led the NLP extraction process using **spaCy** and **GPT-4**.
 - Extracted 18 functional requirements (10 for scheduling/booking, 8 for payment/billing) from the Lab 2 system description.
 - Identified 5 non-functional requirements (e.g., "Payment processing must complete within 2 seconds").
 - Generated an initial completeness report comparing extracted requirements against Lab 3's solution guideline.
- **Bhavneet:**
 - Assisted in refining NLP scripts to handle ambiguous terms (e.g., "real-time availability").
 - Performed manual conflict checks on extracted requirements.
 - Flagged 2 inconsistencies:
 - *Conflict 1:* "Users can cancel bookings up to 1 hour before" vs. "Non-refundable bookings cannot be canceled."
 - *Conflict 2:* "Payments via credit card are instant" vs. "Payments require 5-minute verification."
- **Jaideep:**
 - Automated requirement categorization (functional vs. non-functional) using **NLTK**.
 - Validated extraction accuracy by cross-referencing Lab 2's textual descriptions.
 - Compiled a preliminary list of missing requirements (e.g., "Handling overlapping bookings for same service").

2. Current Responsibilities

Team Member	Tasks	Deliverables
Kanwarjot	NLP extraction finalization, completeness analysis, conflict resolution.	Final list of verified requirements (functional/non-functional).
Bhavneet	LTL property drafting for scheduling & booking , SPIN model setup.	LTL properties for scheduling logic; SPIN model for booking conflicts.
Jaideep	LTL property drafting for payment & billing , SPIN test automation.	LTL properties for payment workflows; SPIN scripts for billing verification.

3. Plan for Next Week (July 19–25)

Kanwarjot

- Finalize requirement extraction by July 20.
- Resolve flagged conflicts through stakeholder assumptions (e.g., "Non-refundable bookings override general cancellation rules").
- Submit verified requirements to Bhavneet/Jaideep for model checking by July 22.

Bhavneet

- Complete 4+ LTL properties for scheduling (e.g., "`G!(booked[user1] && booked[user2])`" for double-booking prevention).
- Build SPIN model for scheduling by July 23.
- Run initial conflict checks (e.g., "Can users book unavailable slots?").

Jaideep

- Draft 4+ LTL properties for payment (e.g., "`F payment_confirmed -> service_accessed`" for payment-service linkage).
- Develop SPIN model for payment workflows by July 23.
- Test edge cases (e.g., "Partial payments fail billing").

Team-Level Goals

- Integrate all SPIN models and run full verification by July 24.

- Draft Sections 1–3 of the final report (requirement summary, extraction analysis, model-checking results).
- Submit progress summary to eClass by July 25.